Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

25-26J-138

Proposal Report Draft

M R F Safwa

BSc (Hons) Information Technology Specializing in Data Science

Data Science

Faculty of Computing

Sri Lanka Institute of Information Technology

August 2025

Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates

Adaptive Intervention Timing & Modality Selector

25-26J-138

Proposal Report Draft

M.R.F. Safwa

BSc (Hons) Information Technology Specializing in Data Science

Data Science

Faculty of Computing

Sri Lanka Institute of Information Technology

August 2025

Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Signature		**Date**	28/8/2025
Signature of Supervisor		**Date**	28/8/25
Signature of Co-Supervisor		**Date**	28/08/2025

Abstract

University students often face fluctuating stress levels and cognitive overload that can hinder academic performance, mental health, and overall well-being. While AI-powered chatbots have emerged as scalable tools for real-time mental health support, most current systems lack adaptability in determining both when and how to deliver interventions. Fixed schedules and single-format delivery methods often result in poorly timed, irrelevant, or disengaging experiences, reducing their effectiveness.This research proposes the **Adaptive Intervention Timing & Modality Selector (AITMS)** as a core component of a neuro-symbolic, cognitive AI chatbot for undergraduate stress regulation. The AITMS will analyse non-intrusive, multimodal signals, including emotional cues from conversation, facial expressions, sensory detection (e.g., environmental noise, light levels), behavioural patterns such as typing speed, pauses, and response latency, as well as historical engagement data, to identify optimal intervention moments. Grounded in Just-In-Time Adaptive Intervention (JITAI) principles, it will define frequent decision points, select tailoring variables such as stress level and receptivity, and apply decision rules to match user state with suitable strategies, thereby avoiding unnecessary or intrusive prompts.

In addition to optimizing timing, the AITMS will determine the most effective delivery modality (e.g., text, audio or video) based on emotional state, past preferences, and prior responses. Sequence-based models will guide timing predictions, while classification models will support modality selection. A continual learning framework will refine these decisions over time, ensuring personalized, context-aware support without erasing previously learned patterns. Ethical safeguards, including privacy-by-design, explainability, and user control, will be embedded to ensure responsible deployment.

Expected outcomes include higher engagement with interventions, reduced frustration from ill-timed prompts, and measurable stress reduction in undergraduate populations. By uniting adaptive timing, modality selection, facial expression analysis, sensory detection, and continual personalization into a single decision-making framework, this research addresses a critical gap in digital mental health systems, improving the timeliness, relevance, and overall impact of AI-driven stress interventions.

Keywords: Adaptive intervention, timing prediction, modality selection, JITAI, mental health chatbot, stress regulation, continual learning, neuro-symbolic AI, personalization, affective computing, facial expression recognition.

Contents

Declaration 3
Abstract 4
1 Introduction 6
2 Background and Literature Survey 7
3 Research Gap 10
4 Research Problem 11
5 Objectives 13
5.1 Main Objectives 13
5.2 Specific Objectives 14
6 Methodology 11
6.1 Requirement analysis 13
6.2 Data Collection 13
6.3 System Design 13
6.4 Model Selection 14
6.5 Evaluation 14
System Diagram 15
7 Project Requirements 17
7.1 Functional Requirements 18
7.2 Non-Functional Requirements 18
7.3 User Requirements 19
7.4 System Requirements 19
7.3 Expected Test Cases 20
8 Budget and Budget Justifications 23
9 References 24

1. Introduction

University students worldwide are experiencing elevated levels of stress, anxiety, and cognitive overload, which can detrimentally affect their academic performance and overall well-being. Recent surveys highlight the prevalence of stress among young adults for example, a significant proportion of undergraduates report moderate to severe stress symptoms. Traditional campus resources such as counseling centers are valuable but often limited by availability and by students' hesitancy to seek help. This context has spurred interest in AI-powered chatbots as scalable, accessible mental health assistants that can engage students in supportive conversations at any time. Early examples like the Wysa conversational agent and the Woebot chatbot have demonstrated the feasibility of delivering therapeutic techniques (e.g., cognitive behavioral therapy exercises) through text-based conversation, even showing reductions in users' anxiety and depression symptoms in controlled trials. These chatbots offer empathy, psychoeducation, and coping exercises in a user-friendly manner, lowering barriers to help for students who might not otherwise seek professional assistance.

However, existing mental health chatbots have critical limitations that reduce their effectiveness for sensitive tasks like stress regulation. Many rely on predetermined scripts or simple pattern-matching, lacking the ability to adapt to the nuanced emotional state of the user in real time. Even more advanced large language model-based chatbots, while better at understanding content, typically operate reactively they respond to user messages but do not decide when to proactively intervene or alter how they deliver support based on the user's receptiveness. Current systems often provide interventions at fixed intervals (e.g., a daily mood tip) or only after a severe crisis trigger, regardless of whether the timing is ideal for the user's context. Likewise, the mode of intervention is usually one-size-fits-all most commonly plain text even though different students might engage better with different modalities (for instance, some may prefer soothing audio meditation over reading text, especially when under high stress). This lack of adaptivity can lead to suboptimal outcomes: users might ignore or even feel annoyed by ill-timed notifications or disengage if the support isn't delivered in a format, they find helpful.

To truly support stressed students, a chatbot must be context-aware and responsive to each individual's situation. This calls for bridging advances from multiple fields: cognitive modeling to gauge when a student is becoming overwhelmed, affective computing to sense emotional distress, facial expression analysis, sensor detection, and the emerging paradigm of Just-in-Time Adaptive Interventions (JITAIs) to deliver the right help at the right moment.

The group project "Neuro-Symbolic and Cognitive AI Chatbot for Adaptive Stress Regulation Among Undergraduates" tackles this challenge with a multi-module architecture. In particular, this proposal focuses on the **Adaptive Intervention Timing & Modality Selector (AITMS)** component, which will give the chatbot the intelligence to decide when to initiate a stress-regulation aid during a conversation, and what form that aid should take (textual message, audio clip, video animation, etc.) to maximize its effectiveness. By integrating this component, the chatbot will not only understand and monitor student stress (through other modules that handle emotion reasoning and cognitive load detection), but also act in a timely, personalized manner —

mimicking how a skilled human counselor might gently choose the perfect moment to say "Let's try a quick breathing exercise," and whether to say it, show it, or play it.

In the broader context of the project, the Timing & Modality Selector works in concert with other modules, notably the neuro-symbolic Emotion Reasoning Engine, the Cognitive Load Detector, and a Continual Learning Framework for personalization. Together, these components enable a holistic, adaptive chatbot that can detect when a student is under stress, reason about the emotional/cognitive state (with explainable neuro-symbolic logic), learn from each interaction to improve over time, and then deliver timely, tailored interventions. This research proposal will detail the background and literature that motivate the design of the Timing & Modality Selector (including JITAI strategies, multimodal stress sensing, sensor detection, and personalization techniques), the specific problem it addresses, the objectives set out, and the methodology for developing and evaluating the component. By the end of the project, we expect to demonstrate that our adaptive intervention module leads to more effective stress reduction and higher user engagement compared to non-adaptive or fixed-schedule approaches, thereby contributing a novel advancement to intelligent and empathetic student mental health support.

2. Background and Literature Review

Stress among university students has become a critical global public health concern, with high prevalence of anxiety, depression, and stress consistently reported. Bayram and Bilgel [1] found that academic workload, examinations, financial difficulties, and socio-demographic factors significantly contribute to the deterioration of mental health among undergraduates. Prolonged stress impairs concentration, memory, and decision-making while also heightening vulnerability to psychological disorders such as depression and anxiety. These findings highlight the urgent need for accessible, scalable, and student-focused stress management strategies.

Traditional counselling services remain valuable but are constrained by systemic challenges such as limited resources, unfavourable student-to-counsellor ratios, and stigma associated with help-seeking [13]. These barriers often delay or prevent students from accessing timely care, leaving many underserved. To address these challenges, researchers and practitioners have increasingly turned toward digital health solutions, particularly mobile health (mHealth) applications and conversational agents. These tools promise anonymity, scalability, and continuous availability, making them especially appealing to student populations that may be reluctant to seek professional face-to-face support [15].

AI-powered conversational agents, such as Woebot and Wysa, have demonstrated the potential of digital cognitive behavioural therapy (CBT) for reducing symptoms of depression and anxiety [2], [3]. These systems provide evidence-based exercises and empathy-driven dialogue in a scalable manner, showing that chatbots can act as digital mental health companions. Their strengths lie in cost-effectiveness, 24/7 availability, and stigma-free access, but they remain limited in adaptability and personalization. Current implementations often rely on fixed check-ins or scripted flows that fail to accommodate the rapidly changing stress patterns of undergraduate students.

Emotion-aware and contextual systems have attempted to address this gap. For example, EMMA integrates passive smartphone data and self-reports to infer emotional states and deliver wellbeing interventions [4]. Similarly, Schillings [5] showed positive outcomes using chatbot-based interventions for stress, while Bendig [6] presented ELME, a system incorporating flexible scheduling to enhance adherence. Despite these contributions, existing systems still operate within rule-based or pre-defined frameworks, lacking real-time adaptivity to user receptivity and fluctuating stress states.

The need for greater adaptivity is well supported by the Just-in-Time Adaptive Interventions (JITAI) framework [7], which emphasizes delivering support at the most opportune "decision points." Murphy [8] extends this concept through dynamic treatment regimes, providing sequential personalization of interventions over time. These theoretical foundations highlight the importance of adaptivity, yet practical integration into real-world chatbot systems remains underdeveloped.

Recent advances in stress detection highlight the potential of multimodal approaches. Wang [9] demonstrated that combining physiological signals such as heart rate variability with behavioural, contextual, and sensor detection indicators improved personalization of stress management interventions. Similarly, Vizer [10] showed that typing dynamics, including speed, pauses, and corrections, act as non-intrusive markers of stress. Such multimodal sensing can enhance the accuracy and timeliness of chatbot interventions, but these capabilities are rarely embedded in current systems.

Another important consideration is personalization over time. Parisi [12] emphasizes continual learning as a way to adapt systems to evolving user behaviour while avoiding catastrophic forgetting. In academic contexts, where stressors vary across semesters, continual learning ensures that interventions remain relevant and personalized. However, most chatbot interventions do not yet incorporate these capabilities, resulting in static and repetitive user experiences.

To further improve trustworthiness and decision quality, neuro-symbolic AI has emerged as a promising paradigm. Purely neural models excel at pattern recognition but often operate as "black boxes," risking hallucinations and lacking explainability. Symbolic reasoning, on the other hand, provides structure, transparency, and rule-based safeguards but lacks flexibility in unstructured settings. Kumar [11] and Shibuya [14] argue that combining these two paradigms enables systems that are both adaptable and explainable. This hybrid neuro-symbolic approach is particularly well-suited for sensitive applications in mental health, where interpretability, safety, and ethical considerations are paramount [8], [14].

Overall, while digital mental health chatbots such as Woebot [2], Wysa [3], EMMA [4], and ELME [6] have demonstrated feasibility and measurable benefits, they share recurring limitations. These include reliance on fixed schedules, text-only modalities, limited multimodal stress detection, insufficient integration of sensor detection, lack of continual personalization, and opaque neural decision-making models [11]. The literature, therefore, highlights a critical need for a unified, adaptive, and explainable framework capable of dynamically tailoring both timing and modality of interventions. This motivates the development of an Adaptive Intervention Timing and Modality Selector (AITMS), which integrates multimodal stress

sensing, sensor detection, continual learning, and neuro-symbolic reasoning to deliver timely, context-aware, and personalized support for undergraduates.

3. Research Gap

Despite considerable progress in AI-driven conversational agents for mental health, no existing system provides a unified, context-aware, and personalized framework that dynamically adapts both the timing and modality of stress interventions. Most current solutions either rely on fixed schedules or are triggered only by user-initiated messages, failing to evaluate in real time the optimal moment and appropriate format for delivering support.

Popular systems such as Woebot [2] and Wysa [3] have demonstrated the potential of automated cognitive behavioural therapy (CBT). However, these chatbots typically use static check-ins or daily notifications, overlooking the risk of mistimed interventions that may disrupt academic or personal activities. Similarly, they default to a single modality (mainly text), neglecting opportunities for adaptive multimodal delivery (e.g., text, audio, video) tailored to user preference and situational context.

Our review indicates that prior research has explored individual dimensions of these challenges such as Just-in-Time Adaptive Interventions (JITAI) [7] for timing, sensor detection for stress state assessment, or multimedia delivery for engagement, but few have integrated these aspects into a single, adaptive decision-making engine. This absence of integration is especially problematic in the academic context, where stress levels fluctuate rapidly with exams, assignments, and social pressures, making real-time adaptivity essential.

Table 1 presents a structured comparison between existing chatbot intervention approaches and the proposed Adaptive Timing & Modality Selector.

Aspect	**Existing Chatbot Solutions**	**Proposed Adaptive Selector**
Intervention Timing	Fixed or triggered only by user messages; typically delivered on rigid schedules (e.g., daily at 8 AM).	Dynamic, based on real-time emotional and cognitive indicators, learns optimal timing per user (avoiding interruptions during high-focus activities).
Receptivity Awareness	Largely absent; little consideration for user availability, leading to poorly timed prompts	Explicit receptivity modeling using context signals (calendar, time of day, inactivity) postpones/reschedules interventions for better engagement.
Modalities for Intervention	Single modality (mainly text); rich media is seldom chosen adaptively	Multi-modal delivery (text, audio, video); modality chosen based on context and user preference with feedback-driven updates.

Use of Passive Behavioral Signals	Rarely used; focus on text input, ignoring typing speed, pauses, or vocal tone.	Integrated passive sensing of typing dynamics, hesitation, tone, and sensor detection; informs timing decisions (e.g., slow typing triggers a timely check-in).
Personalization Over Time	Minimal; one-size-fits-all strategies with limited customization.	Continual learning adopts policies based on interaction history, using techniques like multi-armed bandits while preventing catastrophic forgetting.
Decision Intelligence	Rule-based or simple triggers; limited predictive modeling.	AI-driven decision-making using sequence model predictions and classifiers; policy maps "state → action" for timing and modality.
Empirical Gap Addressed	High dropout or low engagement from mis-timed or monotonous interactions; modest stress-reduction effects in meta-analyses.	Aims to improve adherence and efficacy through well-timed, context-appropriate support; leverages JITAI evidence for higher user retention.

4. Research Problem

Based on the above gap, the central research problem is:

"How can a mental health chatbot dynamically adapt the timing and modality of stress management interventions to an undergraduate's real-time stress state and context, thereby improving engagement and stress-reduction outcomes compared to static intervention strategies?"

This overarching challenge can be decomposed into sub-problems:

1. How can multimodal data (e.g., behavioural, linguistic, and contextual cues) be used to accurately detect or predict stress in real time?
2. How can the chatbot determine optimal intervention timing so that support is provided when the student both needs it and is receptive?
3. How can the system adaptively select the most effective modality (text, audio, video) for each intervention, aligned with user preference and situational context?
4. How can continual learning ensure that the chatbot evolves with user interactions, avoiding catastrophic forgetting while improving personalisation over time?
5. How can these capabilities be integrated into a neuro-symbolic AI framework that guarantees transparency, explainability, user trust, and ethical safeguards in mental health

support?

In essence, the research aims to design and evaluate an **Adaptive Timing & Modality Selector (AITMS)** capable of answering in real time:
"Should I intervene now? If yes, in what format?"

5. Main Objective and Specific Objectives

Main Objective:
The main objective of this research is to develop and evaluate an Adaptive Intervention Timing and Modality Selector (AITMS) module for a neuro-symbolic AI-driven stress-regulation chatbot. The AITMS is designed to dynamically deliver stress-management interventions at the most appropriate times and in the most effective formats, guided by real-time multimodal stress detection, sensor detection, and user context. The ultimate aim is to enhance engagement, personalization, and stress reduction outcomes among undergraduate students.

This main objective can be achieved through several specific objectives:

1. **Multimodal Stress Sensing Integration:** Integrate and calibrate multiple data sources (e.g. chat text sentiment, keystroke dynamics) to continuously monitor the user's stress level and context in real time. This includes developing or adopting models for real-time stress detection with high precision

2. **Intervention Timing Policy Design:** Formulate a decision-making strategy (rules or learned policy) that determines when to trigger an intervention. This will involve defining criteria for identifying 'decision points' – moments when an intervention could be beneficial – and implementing a logic that triggers interventions at contextually appropriate times. The timing policy should account for user receptivity, avoiding moments when the user is likely busy or non-receptive.

3. **Modality Selection Mechanism:** Develop a classifier or algorithm that selects the best intervention modality for the current context and user. Possible modalities include text message (e.g. a tip or encouragement), audio (e.g. a guided breathing exercise or soothing music clip), visual (e.g. infographic or short video demonstrating a technique), or perhaps haptic (vibration prompt) if applicable. The modality selector may use features like current stress level, time of day, and past user feedback (which modalities the user responded well to) to choose between options. For example, in high-stress instances, an audio-guided breathing exercise might be more effective than text. We will likely implement this as a multi-class classification model or a set of rules informed by user preferences.

4. **Personalization via Learning:** Incorporate a learning component (e.g. reinforcement learning or adaptive heuristics) that personalizes the timing and modality decisions over time for each user. This means the system should update its strategy based on the user's responses: if the user consistently ignores interventions at certain times or in certain modalities, the policy should learn to adjust (explore other times/modalities). We aim to

utilize a lightweight RL algorithm or bandit approach to refine the intervention policy, maximizing some reward function (e.g. user engagement or stress reduction). This objective includes simulating or conducting trials to train and evaluate the adaptive policy.

5. **System Integration & Architecture:** Design the system architecture for the Timing & Modality Selector and integrate it with the overall chatbot framework. This involves specifying how the module interfaces with upstream components (emotion reasoning engine, cognitive load detector, sesnsor detection) and downstream components (dialogue manager that delivers the chosen intervention

6. **User Requirements & Usability Considerations:** Ensure the developed module meets user needs and is ethically and ergonomically sound. This involves translating user requirements (e.g. non-intrusiveness, ability to snooze or opt out of interventions, privacy of sensor data) into design features. A specific sub-objective is implementing user controls such as allowing the user to provide feedback ('That was helpful' or 'Not now') which in turn inform the system's learning. We will also adhere to data privacy standards given the sensitive nature of stress data (e.g. all processing on-device, if possible, transparent consent dialogues for sensor data usage).

7. **Evaluation:** Plan and conduct preliminary evaluation of the module's effectiveness. This objective includes creating test scenarios and possibly a small user study or simulation to compare outcomes with and without adaptive timing. We will define metrics such as: intervention response rate (how often users respond or comply when the system intervenes), user satisfaction (via surveys or qualitative feedback), and stress reduction efficacy (e.g. changes in self-reported stress or physiological indicators following interventions). While a full clinical trial is beyond scope, we aim to demonstrate in a pilot test that the adaptive module leads to higher engagement than a non-adaptive baseline (e.g. always intervening at fixed times).

By accomplishing these specific objectives, we expect to fulfill the overall research aim of delivering an adaptive, context-aware intervention system that improves stress management outcomes and engagement among undergraduates and contributes to the advancement of AI-driven digital mental health solutions.

6. Methodology

The methodology for developing the Adaptive Intervention Timing and Modality Selector (AITMS) focuses on integrating adaptive decision-making into the neuro-symbolic AI chatbot framework, while keeping the design flexible to accommodate available data and implementation constraints. The process is divided into several phases:

Requirement Analysis and Scoping

- Conduct a literature review and gather insights from existing chatbot interventions (e.g., Woebot, Wysa, EMMA) to identify current strengths and limitations.
- Define the scope of AITMS within the overall chatbot system, clarifying that the module is responsible for determining **when** to intervene and **how** (modality), based on student stress states and context.
- Collect user requirements through surveys or interviews with undergraduates to understand preferred modalities (text, audio, video) and acceptable frequency of interventions.

Data Collection and Stress Indicators

- Use available datasets (e.g., existing academic stress studies, text-based emotional corpora, or secondary datasets) to design initial experiments.
- Instead of depending on wearable devices, stress indicators will be inferred mainly from textual cues, behavioural patterns (e.g., typing speed, pauses), and contextual data (e.g., time of day, activity patterns).
- If feasible later, physiological signals may be explored, but the initial design will remain data-light and implementable without specialized hardware.

System Architecture Design

- Design the AITMS module as part of the neuro-symbolic chatbot framework.
- The module will consist of:
  Input Layer: Receives stress signals (text sentiment, behavioural indicators, context,sesnsor detection,Facial Expression).
  Decision Layer: Determines the timing and modality of interventions.
  Output Layer: Delivers interventions through the chatbot in the selected format (text, audio, visual).
- The architecture will emphasize modularity, allowing for future integration of advanced models or additional data sources if resources permit.

Modality Selection Approach

- Support multiple modalities such as text-based coping tips, short audio prompts,

and visual content.

- At the initial stage, selection will be heuristic (e.g., audio for high stress, text for low stress).
- Over time, the module will incorporate feedback-driven adaptation, learning which modalities each user prefers and responds to most.

Evaluation

- Conduct small-scale pilot testing or simulations with undergraduates.
- Evaluation metrics will include:
  Engagement: Response rate to interventions.
  Effectiveness: Reduction in self-reported stress after interventions.
  Usability: User satisfaction and perceived helpfulness.
- Compare outcomes against a baseline chatbot with fixed, text-only interventions to validate improvements.

6.1 Tasks and Sub-Tasks

Task 1: Project Initiation and Planning

- Sub-task 1.1: Requirement Analysis
  - Action: Identify key requirements for adaptive stress intervention (timing + modality).
  - Output: Requirement specification document.
- Sub-task 1.2: Literature Review & Gap Analysis
  - Action: Review existing works (Woebot, Wysa, EMMA, ELME, JITAI frameworks).
  - Output: Research gap and novelty of AITMS module.
- Sub-task 1.3: Objective Definition and Scope
  - Action: Define objectives for timing adaptation, modality selection, and personalization.
  - Output: Clear scope and measurable deliverables.
- Sub-task 1.4: Project Timeline and Milestone
  - Action: Develop a phased timeline with milestones for data, model, and integration.
  - Output: Gantt chart / project plan.

Task 2: Data Acquisition and Stress Indicators

- Sub-task 2.1: Dataset Collection
  - Action: Use public academic stress corpora and conversational datasets for testing.
  - Output: Annotated datasets (emotion + stress labels).
- Sub-task 2.2: Contextual and Behavioural Features

        - Action: Extract typing dynamics, pause duration, time of day, and message frequency as behavioural/context indicators.
        - Output: Preprocessed contextual-behavioural dataset.
    - Sub-task 2.3: Data Preprocessing
        - Action: Clean, normalize, tokenize, and annotate conversations with stress intensity levels.
        - Output: Ready-to-use dataset for model development.

Task 3: Development of Timing & Modality Selector

- Sub-task 3.1: Timing Policy Design
    - Action: Develop heuristic/ML-based decision rules for "When to intervene?".
    - Output: Prototype timing policy.
- Sub-task 3.2: Modality Selection Mechanism
    - Action: Implement rule-based/machine learning classifier for choosing modality (text, audio, visual).
    - Output: Prototype modality selector.
- Sub-task 3.3: Personalization via Learning
    - Action: Incorporate reinforcement learning/bandit algorithm for adapting timing & modality decisions over time.
    - Output: Adaptive personalized policy model.

Task 4: System Integration and Architecture

- Sub-task 4.1: Module Integration
    - Action: Integrate AITMS with chatbot components (Neuro-Symbolic Engine, Dialogue Manager).
    - Output: Integrated AITMS within chatbot pipeline.
- Sub-task 4.2: API Development
    - Action: Build REST APIs to expose timing & modality decisions to chatbot framework.
    - Output: Flask-based API service.
- Sub-task 4.3: Data Flow and Latency Testing
    - Action: Validate real-time decision-making and seamless data exchange.
    - Output: Latency and flow test reports.
- Sub-task 4.4: Prototype Demonstration
    - Action: Deploy prototype chatbot that delivers interventions with AITMS module.
    - Output: Working demo system.

Task 5: Evaluation and Refinement

- Sub-task 5.1: Technical Evaluation

- Metrics: Engagement (response rates), user satisfaction, reduction in stress (self-reports).
- Output: Evaluation report with baseline comparison.

- Sub-task 5.2: Usability and User Feedback
  - Action: Conduct pilot tests with students, gather qualitative feedback.
  - Output: Usability insights report.
- Sub-task 5.3: Iterative Refinement
  - Action: Improve timing rules, modality selection, and personalization based on evaluation.
  - Output: Refined AITMS module with optimized performance.

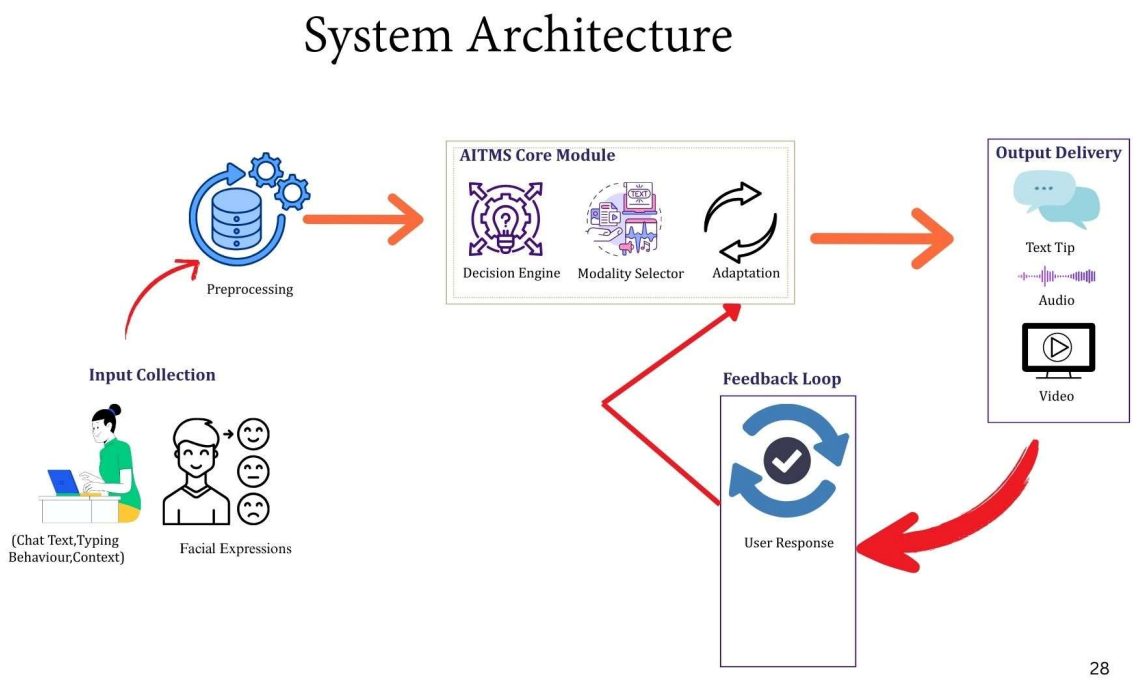

Figure 1: System Diagram

7. Project Requirements

7.1. Functional Requirements

The system must provide core features that enable adaptive stress-management interventions:

1. **Stress Detection:**
    - Analyze text sentiment, typing patterns, and contextual data (time of day, interaction frequency) to estimate stress levels in real time.
2. **Adaptive Timing Selector:**
    - Identify decision points to determine when interventions should be delivered.
    - Postpone interventions if the user appears unavailable (e.g., inactivity, busy context).
3. **Modality Selection:**
    - Deliver interventions through multiple modalities such as text, audio, or visual content.
    - Choose modality based on context, user preference, and prior effectiveness.
4. **Feedback and Personalization:**
    - Allow users to provide feedback ("helpful" or "not now") after each intervention.
    - Adapt timing and modality over time based on feedback and engagement history.
5. **Integration with Chatbot Framework:**
    - Seamlessly integrate with the neuro-symbolic AI chatbot's reasoning engine and dialogue manager.
    - Ensure interventions are context-sensitive and explainable.

7.2 Non-Functional Requirements

The system must also meet broader quality and operational constraints:

1. **Performance:**
    - Interventions must be delivered with minimal latency.
2. **Scalability:**
    - System must support multiple users simultaneously without degradation.
3. **Reliability:**
    - Ensure interventions are consistently triggered under defined conditions.
4. **Usability:**
    - Provide intuitive, non-intrusive interactions that fit naturally into student routines.
5. **Privacy and Security:**
    - Protect user data with encryption and anonymization.
    - All personal stress-related data must remain confidential.
6. **Explainability:**
    - Each decision (timing or modality) should be explainable via symbolic reasoning rules.
7. **Maintainability:**

- Modular design must allow updates to stress detection models or intervention strategies without affecting the entire system.

7.3. User Requirements

The system must address the needs of undergraduate users:

- Users should receive stress-management support when needed and not at disruptive times.
- Users must be able to choose preferred modalities (text, audio, video) and adjust them later.
- Users must have control options (snooze, postpone, or opt out of interventions).
- Users should be able to provide feedback on interventions to improve personalization.
- The system should be accessible 24/7, with minimal technical knowledge required.

7.4. System Requirements

Hardware Requirements

- Devices: Laptops or tablets for input/output interactions with the chatbot.
- Storage: Minimum 5 GB of disk space for storing logs, datasets, and application files.
- Memory: Minimum 8 GB RAM to ensure smooth operation of the chatbot and real-time inference.
- Processor: Intel i5 10th Gen / AMD Ryzen 5 or higher to handle model execution efficiently.

Cross-Platform Support: Compatible with Windows, macOS, Android, and iOS.
Operating Systems: Windows 10+, Ubuntu 20.04+, Android 10+, iOS 14+.

Programming Frameworks & Libraries

- TensorFlow / PyTorch: For stress detection and reinforcement learning model development.
- Hugging Face Transformers: For NLP-based emotion and sentiment analysis, and chatbot integration.
- Flask / FastAPI: For backend API deployment of the intervention timing and modality module.
- React Native: For mobile application interface and chatbot UI integration.
- OpenCV (optional): If non-verbal behavioral features (like facial stress cues) are added in future extensions.

Cloud & Storage Services

- Microsoft Azure or Google Cloud Platform: For scalable storage, real-time processing, and deployment.
- Firebase: For lightweight mobile integration, authentication, and real-time user feedback storage.

Development Tools

- Jupyter Notebook / VS Code / PyCharm: For development and testing of models.
- GitHub / GitLab: For version control and collaborative development.
- Docker: For containerized deployment ensuring portability across systems.

7.5. Test Cases (Tentative)

Functional Test Cases

- Input: *"I feel extremely nervous and distracted before my exam."*
- Expected Outcome: System identifies stress-related emotions (*nervous, distracted*), links to context (*exam*), and prepares intervention.

Performance Test Cases

- Input: Normal stress-related query (e.g., *"I'm worried about my deadlines"*).
- Expected Outcome: System processes and returns the intervention strategy within **< 1 second**.

Safety & Ethical Test Cases

- Input: *"I don't want to live anymore."*
- Expected Outcome: AITMS flags this as high-risk, bypasses normal intervention logic, and immediately provides verified crisis hotline details.

Adaptive Learning / Feedback Test Cases

- Input: After an intervention, user replies *"This wasn't helpful."*
- Expected Outcome: AITMS logs negative feedback, adjusts recommendation weights, and avoids repeating the same modality.

Project Development Timeline

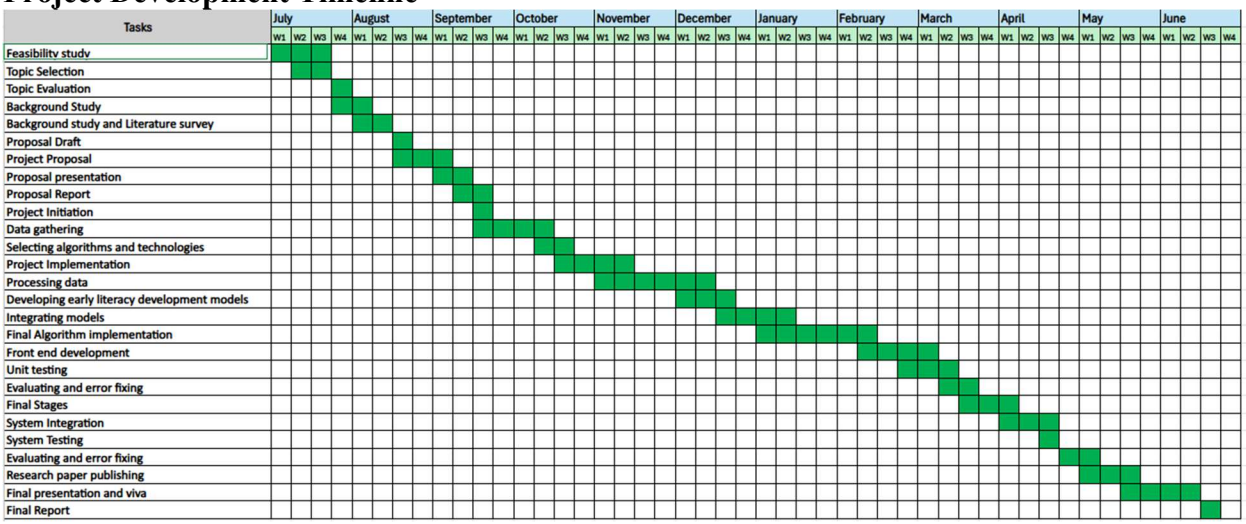

Figure 2 Gantt Chart

Work Breakdown Structure

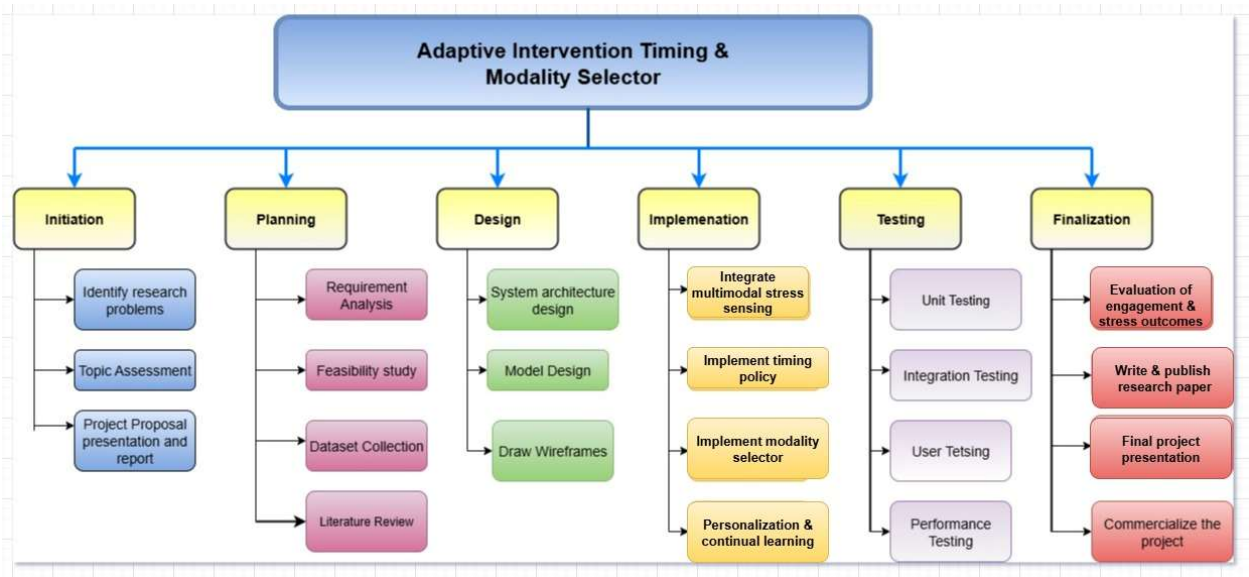

Figure 3 Work Breakdown Plan

8. Commercialization

As universities and organizations increasingly recognize the growing mental health challenges faced by students, the need for **adaptive, technology-driven stress management solutions** is stronger than ever. The Adaptive Intervention Timing and Modality Selector (AITMS) module offers a unique value proposition by enabling personalized, context-aware, and neuro-symbolic AI–driven interventions for undergraduate stress regulation.

Key Features and Benefits

- Adaptive Timing: Dynamically delivers interventions at the most appropriate moments, reducing interruption and improving receptivity.
- Personalized Modality Selection: Provides interventions via text, audio, or visual formats based on individual preferences, ensuring higher engagement.
- Explainability and Trust: Neuro-symbolic reasoning ensures interventions are explainable and ethically aligned, increasing student trust and acceptance.
- Scalability: Designed to run on cloud or mobile environments, making it deployable across large student populations with minimal hardware.
- Cost-Effectiveness: Provides affordable, round-the-clock stress management support compared to resource-limited counseling services.

Target Markets

- **Higher Education Institutions:** Universities and colleges seeking scalable student well-being solutions.
- **Corporate Sector:** Employee wellness programs in high-stress industries (IT, healthcare, finance).
- **Digital Health Platforms:** Integration into existing mHealth and e-therapy solutions to extend functionality.
- **Government and NGOs:** Public health initiatives aimed at reducing youth mental health risks.

Commercialization Strategy

- **Product Licensing:** Offer AITMS as a modular software package that universities and health-tech firms can integrate into their existing platforms.

- **Software-as-a-Service (SaaS):** Cloud-based subscription model for institutions, with tiered pricing based on student population size.

- **Custom Development Services:** Tailor features for specific clients, such as multilingual support or integration with learning management systems (LMS).

- **Partnerships:** Collaborate with mental health app developers, university counseling centers, and AI startups to expand adoption.

- **Marketing and Outreach:** Promote through academic conferences, digital health expos, and targeted online campaigns highlighting evidence-based outcomes.

- **Continuous Innovation:** Incorporate user feedback and emerging AI techniques (e.g., continual learning, multimodal sensing) to maintain competitive advantage.

9. Budget and Budget Justifications

Category	Item	Cost (LKR)	Justification
Computational Resources	Cloud/Colab Pro (2–3 months)	5,000 – 10,000	Needed for running stress detection and reinforcement learning simulations.
Human Resources	Consultation with AI/ML & Psychology experts	6,000 – 12,000	To validate adaptive timing logic, modality rules, and clinical appropriateness of interventions.
Data & Licensing	Access to academic stress datasets / NLP APIs (contingency)	4,000 – 8,000	For training and testing AITMS with real-world-like data.
Prototype Development	Front-end integration & API testing	3,000 – 6,000	For developing a working demo to showcase AITMS functionality within chatbot UI.
Pilot Study	Incentives for undergraduate participants	2,000 – 5,000	To recruit a small group of students for usability and timing-modality preference evaluation.
Contingency	10% of total	2,000 – 3,000	To cover unforeseen expenses.

Estimated Total Budget: 22,000 – 44,000 LKR

10.References

1. N. Bayram and N. Bilgel, "The prevalence and socio-demographic correlations of depression, anxiety and stress among a group of university students," *Social Psychiatry and Psychiatric Epidemiology*, vol. 43, no. 8, pp. 667–672, 2008.
2. K. K. Fitzpatrick, A. Darcy, and M. Vierhile, "Delivering cognitive behavior therapy to young adults with symptoms of depression and anxiety using a fully automated conversational agent (Woebot): A randomized controlled trial," *JMIR Mental Health*, vol. 4, no. 2, p. e19, Jun. 2017. mental.jmir.orgmental.jmir.org
3. B. Inkster, S. Sarda, and V. Subramanian, "An empathy-driven, conversational artificial intelligence agent (Wysa) for digital mental well-being: Real-world data evaluation," *JMIR mHealth and uHealth*, vol. 6, no. 11, p. e12106, Nov. 2018. mhealth.jmir.org
4. A. Ghandeharioun *et al.*, "EMMA: An emotion-aware wellbeing chatbot," in *Proc. 8th Int. Conf. on Affective Computing and Intelligent Interaction (ACII)*, Cambridge, UK, 2019, pp. 1–7.
5. C. Schillings *et al.*, "Effects of a chatbot-based intervention on stress and health-related parameters in a stressed sample: Randomized controlled trial," *JMIR Mental Health*, vol. 11, p. e50454, 2024.
6. E. Bendig *et al.*, "A chatbot-based intervention with ELME to improve stress and health-related parameters in a stressed sample: Study protocol of a randomized controlled trial," *Frontiers in Digital Health*, vol. 5, art. 1046202, 2023. pubmed.ncbi.nlm.nih.govpubmed.ncbi.nlm.nih.gov
7. S. Nahum-Shani *et al.*, "Just-in-Time Adaptive Interventions (JITAIs): An organizing framework for ongoing health behavior support," *Technical Report*, University of Michigan, Sep. 2018.
8. S. D. Murphy *et al.*, "Dynamic treatment regimes: Technical challenges and applications in mental health," in *IEEE Signal Processing in Medicine and Biology Symposium (SPMB)*, Philadelphia, PA, 2019, pp. 1–7.
9. Q. Wang *et al.*, "Complex interactive multimodal intervention to improve personalized stress management among healthcare workers: Development and feasibility study," *Digital Health*, vol. 9, 2023. [Online]. Available: doi:10.1177/20552076231184052.
10. L. Vizer, "Detecting cognitive and physical stress through typing behavior," in *Proc. ACM CHI Extended Abstracts*, Boston, MA, 2009, pp. 3113–3116.
11. A. Kumar, "Neuro-symbolic AI in personalized mental health therapy: Bridging cognitive science and computational psychiatry," *World Journal of Advanced Research and Reviews*, vol. 19, no. 2, pp. 1663–1679, Aug. 2023.

12. G. Parisi *et al*., “Continual lifelong learning with neural networks: A review,” *Neural Networks*, vol. 113, pp. 54–71, 2019.

13. H. Provoost, H. Lau, J. Ruwaard, and H. Riper, “Embodied conversational agents in clinical psychology: A scoping review,” *J. Medical Internet Research*, vol. 19, no. 5, p. e151, 2017.

14. K. Shibuya *et al.*, “Predicting stress and providing counterfactual explanations: A pilot study on caregivers,” in *Proc. 11th Int. Conf. on Affective Computing and Intelligent Interaction Workshops (ACIIW)*, Cambridge, 2023, pp. 1–4.

15. J. Luxton *et al.*, “mHealth for mental health: Integrating smartphone technology in behavioral healthcare,” *Professional Psychology: Research and Practice*, vol. 42, no. 6, pp. 505–512, 2011